1. Some inequalities for Gaussian measures and the long-range order of the one-dimensional plasma

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1.1. Introduction

THE following is a preliminary report on some recent work, the full details of which will be published elsewhere. We have come across some inequalities about integrals and moments of log concave functions which hold in the multidimensional case and which are useful in obtaining estimates for multidimensional modified Gaussian measures. By making a small jump (we shall not go into the technical details) from the finite to the infinite dimensional case, upper and lower bounds to certain types of functional integrals can be obtained. As a non-trivial application of the latter we shall, for the first time, prove that the one-dimensional one-component quantum-mechanical plasma has long-range order when the interaction is strong enough. In other words, the Wigner lattice can exist, in one dimension at least. As another application we shall prove a log concavity theorem about the fundamental solution (Green's function) of the diffusion equation.

1.2. Basic concavity theorem

We begin with a theorem (Theorem 1.1) which, to the best of our knowledge, is new and which constitutes the basis of all our other inequalities.

DEFINITION 1.1. A function F from $\mathbf{R}^n$ to $\mathbf{R}$ is a *log concave function* if $F(x) \geqslant 0$, $\forall x \in \mathbf{R}^n$, and $F(x)^{\lambda} F(y)^{1-\lambda} \leqslant F[\lambda x + (1-\lambda)y]$, $\forall x, y \in \mathbf{R}^n$ and $\lambda \in (0, 1)$. If the inequality is reversed, we say that F is *log convex*. We shall sometimes write $F(x) = e^{f(x)}$ and f is concave, but it then is understood that f can take on the value $-\infty$. We say that F is *even* if $F(x) = F(-x)$, $\forall x$.

Two important examples of log concave functions are:

(a) $F(x) = \exp[-(x, Ax)]$, where A is any symmetric real positive-semidefinite quadratic form on $\mathbf{R}^n$.

(b) Let C be any convex set in $\mathbf{R}^n$ and let $\chi_C(x) = 1$ for $x \in C$, $\chi_C(x) = 0$ for $x \notin C$ be the characteristic function of C. Then χ_C is a log concave function. χ_C is even if and only if C is *balanced*, i.e. $x \in C \Rightarrow -x \in C$.

THEOREM 1.1. *Let F be a log concave function on $\mathbf{R}^{m+n}$ and F: $(x, y) \mapsto F(x, y)$ for $x \in \mathbf{R}^m$, $y \in \mathbf{R}^n$. Then $G(x) \equiv \int_{\mathbf{R}^n} F(x, y)\, dy$ is a log concave function on $\mathbf{R}^m$.*

We have four different proofs of this theorem, one of which is the following.

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Proof. It is sufficient to prove the theorem when $m = n = 1$; the general case follows by Fubini's theorem and induction. Choose two points x and x' such that $G(x) \neq 0$ and $G(x') \neq 0$. We may assume that

$$\sup_y \{F(x, y)\} = \sup_y \{F(x', y)\},$$

for otherwise we can replace $F(x, y)$ by $e^{bx}F(x, y)$ with b suitably chosen. For each $z \geqslant 0$, define

$$C(z) = \{(x, y) | F(x, y) \geqslant z\} \subset \mathbf{R}^2,$$

$C(x, y) = \{y | F(x, y) \geqslant z\} \subset \mathbf{R}$ and $g(x, z) = \text{meas}\{C(x, z)\}$.
Then

(*i*) $C(z)$ is convex and thus $C(x, z)$ is an interval;
(*ii*) $G(x) = \int_0^\infty g(x, z)\, dz$;
(*iii*) for all $0 \leqslant \lambda \leqslant 1$, $g(\lambda x + (1-\lambda)x', z) \geqslant \lambda g(x, z) + (1-\lambda)g(x', z)$.

This last fact follows easily from the convexity of $C(z)$; it is also the Brunn–Minkowski theorem which, in one dimension, is trivial. Thus

$$G(\lambda x + (1-\lambda)x') \geqslant \lambda G(x) + (1-\lambda)G(x') \geqslant G(x)^\lambda G(x')^{1-\lambda}.$$

Q.E.D.

Theorem 1.1 should not be confused with the following theorem, which is much simpler and which follows directly from Hoelder's inequality.

THEOREM 1.2. *Let* $F: \mathbf{R}^{m+n} \to \mathbf{R}$ and, for $x \in \mathbf{R}^m$, $y \in \mathbf{R}^n$, *let* $F(x, y)$ *be log convex in* x *for each fixed* y. *Then* $G(x) \equiv \int_{\mathbf{R}^n} F(x, y)\, dy$ *is log convex on* $\mathbf{R}^m$.

An immediate consequence of Theorem 1.1 is the following.

THEOREM 1.3. *The convolution of two log concave functions on* $\mathbf{R}^n$ *is log concave.*

Proof. $H(x) = \int_{\mathbf{R}^n} F(x-y)G(y)\, dy$ is log concave since $F(x-y)G(y)$ is jointly log concave in $(x, y) \in \mathbf{R}^{n+m}$.

Q.E.D.

REMARK. In the case of $\mathbf{R}$, Theorem 1.3 is known [1].

1.3. Application of Theorem 1.1 to Gaussian measures

A Gaussian measure on $\mathbf{R}^n$ is given by an (unnormalized) density function $W(x) = \exp[-(x, Ax)/2]$, $A > 0$. The expectation value of a real-valued function H, on $\mathbf{R}^n$, is given by

$$\langle H \rangle_0 = \frac{\int H(x) W(x)\, dx}{\int W(x)\, dx}.$$

Now suppose that $W(x)$ is replaced by $W_F(x) = W(x)F(x)$, where F is a log concave function. With respect to the new weight we define $\langle H\rangle_F$ as above. How does $\langle H\rangle_F$ compare with $\langle H\rangle_0$?

THEOREM 1.4. *The covariance matrix M^F, whose elements are $M^F_{ij} = \langle x_i x_j\rangle_F - \langle x_i\rangle_F\langle x_j\rangle_F$, satisfies*

$$M^F \leqslant M^0 = A^{-1}$$

in the sense of forms, i.e. $M^0 - M^F$ is positive-semidefinite.

Proof. Consider the function T on $\mathbf{R}^{n+m}$ defined by

$$\begin{aligned} T(x, y) &= W(x)F(x)\exp[-(y, A^{-1}y)/2 + (y, x)] \\ &= W(x - A^{-1}y)F(x). \end{aligned}$$

Then T is log concave and, by Theorem 1.1, $U(y) = \int dx T(x, y)$ is log concave on $\mathbf{R}^n$. Thus, the matrix $\partial^2 \ln U(y)/\partial y_i \partial y_j|_{y=0} = M^F - M^0$ is negative-semidefinite.

Q.E.D.

THEOREM 1.5. *If, in the above, we replace F by a log convex function then $M^F \geqslant M^0 = A^{-1}$.*

Proof. Write $U(y) = \int_{\mathbf{R}^n} W(x)F(x + A^{-1}y)\,dx$ and use Theorem 1.2.

Q.E.D.

As an application of Theorem 1.4, consider an Ising model with Boltzmann factor $B(\sigma) = \exp[\frac{1}{2}(\sigma, A\sigma)]$, $\sigma_i = \pm 1$, $i = 1, \ldots, n$. By adding an unimportant multiple of the identity to A, we can always assume $A > 0$. Since

$$B(\sigma) = (2\pi)^{-n/2}[\det A]^{-\frac{1}{2}} \int_{\mathbf{R}^n} \exp[-(x, A^{-1}x)/2 + (x, \sigma)]\,dx$$

we find that the covariance matrix of the σs, $N_{ij} \equiv \langle \sigma_i \sigma_j\rangle_B$, is simply related to the covariance matrix M^F, introduced above (with A replaced by A^{-1}), by

$$N = A^{-1}M^F A^{-1} - A^{-1},$$

where

$$F(x) = \sum_\sigma e^{(x,\sigma)} = \prod_{i=1}^n 2\cosh x_i.$$

Now $F(x)$ is log convex, so Theorem 1.5 states that $M^F \geqslant A$, which implies that $N \geqslant 0$—hardly an interesting result. Note, however, that

$$G(x) \equiv F(x)\exp\left(-\frac{1}{2}\sum_{i=1}^n x_i^2\right)$$

is log concave. Therefore, provided $A^{-1} > I$ (equivalently, $A < I$) we can write

$$\exp[-(x, A^{-1}x)/2]F(x) = \exp[-(x, (A^{-1}-I)x)/2]G(x)$$

and Theorem 1.4 states that $M^F \leqslant (A^{-1}-I)^{-1}$ and $N \leqslant (I-A)^{-1}$. In the physical situation, A is a matrix whose eigenvalues are of O(1) independent of n and $A < I$ occurs for sufficiently high temperature, independently of n. Hence, for high temperature, the eigenvalues of N are O(1); this means there is no long-range order. Although previously there existed elementwise bounds on N for special choices of A ([2] and [3]: inequalities), our result is the first case of a quadratic form inequality on N.

We now quote an assortment of theorems, to indicate some of the directions in which Theorem 1.4 can be generalized.

THEOREM 1.6. *Consider the weight $W_F(x) = W(x)F(x)$ with F log concave, as in Theorem 1.4, and let F be even. Let L be any symmetric, real, n-square matrix. Then*

$$\langle (x, Lx)^2 \rangle_F - \langle (x, Lx) \rangle_F^2 \leqslant 2\langle (x, LA^{-1}Lx) \rangle_F. \tag{1.1}$$

Proof. We consider the case in which $A = I$; the general case can be handled by the change of variables $x \to A^{-\frac{1}{2}}x$. Let $Z = \int_{\mathbf{R}^n} e^{-(x,x)/2}F(x)\,dx$. Then

$$2\Delta \equiv 2\langle (x, Lx)^2 \rangle_F - 2\langle (x, Lx) \rangle_F^2$$

$$= Z^{-2} \int_{\mathbf{R}^n} \int_{\mathbf{R}^n} e^{-[(x,x)+(y,y)]/2} F(x)F(y)[(x, Lx) - (y, Ly)]^2\,dx\,dy$$

$$= Z^{-2} \int_{\mathbf{R}^n} \int_{\mathbf{R}^n} e^{-[(u,u)+(v,v)]/2} F[2^{-\frac{1}{2}}(u+v)]\, F(2^{-\frac{1}{2}}(u-v))4(u, Lv)^2\,du\,dv$$

after the change of variables $x = 2^{-\frac{1}{2}}(u+v)$, $y = 2^{-\frac{1}{2}}(u-v)$. Now do the v integration and recall that $\{\langle v_i, v_j \rangle\}_{i,j=1}^n \leqslant I$ for each u, by Theorem 1.4. Thus, $2\Delta \leqslant 4\langle (u, L^2u) \rangle$. Returning to the original x, y variables, one notes that $2(u, L^2u) = (x, L^2x) + (y, L^2y) + 2(x, L^2y)$.

Finally $\langle (x, L^2x) \rangle = \langle (y, L^2y) \rangle = \langle (x, L^2x) \rangle_F$ and $\langle x_i y_j \rangle = 0$.

Q.E.D.

REMARKS. (*i*) If F is log convex, the inequality in Theorem 1.6 is reversed.

(*ii*) The significance of Theorem 1.6 is that if L and A are of the order of I, the left side of (1.1) is the difference of two terms of $O(n^2)$, while the right side is $O(n)$. Choosing $L = A$, the left side of (1.1) is like n times a specific heat, while the right side is like n times an internal energy—to use the language of statistical mechanics. Usually, it is difficult to obtain an upper bound on a specific heat.

COROLLARY 1.7. *Let A and L be symmetric, n-square matrices with A non-singular, let F be even and log concave and let λ be real. Then*

$$Z(\lambda) \equiv \int_{\mathbf{R}^n} \exp[-(x, A\, e^{\lambda L} A x)] F(x)\, dx$$

is log concave in λ.

Proof. Compute $d^2 \ln Z/d\lambda^2$ and compare with Theorem 1.6.

Q.E.D.

THEOREM 1.8. *Let $W_F(x) = e^{-x^2/2} F(x)$ be a weight in $\mathbf{R}$ with F log concave. Define $\langle\, . \,\rangle_F$ and $\langle\, . \,\rangle_0$ as before. Then*

$$\langle |x - \langle x \rangle_F|^\alpha \rangle_F \leq \langle |x - \langle x \rangle_0|^\alpha \rangle_0$$

for $\alpha \geq 1$.

The proof of Theorem 1.8 is lengthy and will not be given here. The theorem says that multiplying a Gaussian weight on $\mathbf{R}$ by a log concave function may, if the function is not even, shift the mean, but all moments, higher than the first, with respect to the new mean are decreased.

We present next a theorem which will play an important role in the next section.

THEOREM 1.9. *Let A be a real positive-definite $(n+m)-$square matrix partitioned as $A = \begin{bmatrix} \alpha & \beta \\ \beta^T & \gamma \end{bmatrix}$, where α is n-square, γ is m-square, β is $n \times m$, and T means transpose. Let F be a log concave function on $\mathbf{R}^{n+m}$ and form the unnormalized weight on $\mathbf{R}^{n+m}$: $W_F(x) = W(x)F(x)$, $W(x) = \exp[-(x, Ax)/2]$. Denoting, as before, a point $x \in \mathbf{R}^{m+n}$ as $x = (y, z)$, $y \in \mathbf{R}^n$, $z \in \mathbf{R}^m$, define the unnormalized weight V on $\mathbf{R}^n$ by*

$$V(y) = \int_{\mathbf{R}^m} W_F(y, z)\, dz.$$

If we define G: $\mathbf{R}^n \to \mathbf{R}$ by

$$V(y) = \exp[-(y, By)/2] G(y),$$

with $B = \alpha - \beta\gamma^{-1}\beta^T > 0$, then G is log concave.

Proof. Note that the $(n+m)$-square matrix $C \equiv A - \begin{bmatrix} B & 0 \\ 0 & 0 \end{bmatrix}$ is positive-semidefinite, since A is positive-definite. Hence $U_F(x) = \exp[-(x, Cx)/2] F(x)$ is log concave on $\mathbf{R}^{n+m}$. Since

$$V(y) = \exp[-(y, By)/2] \int_{\mathbf{R}^m} U_F(y, z)\, dz,$$

Theorem 1.9 follows from Theorem 1.1.

Q.E.D.

REMARKS. (*i*) *Mutatis mutandis*, if F is replaced by a log convex function, then G is log convex on $\mathbf{R}^n$.

(*ii*) If $F(x)$ is a constant, then $G(y)$ is also a constant. Thus, Theorem 1.9 states that if one does a partial integration over a Gaussian weight times a log concave function, the result is the Gaussian weight one would have obtained without the log concave multiplier times a new log concave function.

To pursue the ideas of Theorem 1.9 a bit further, let us formulate the Brunn–Minkowski theorem for Gaussian measures. We recall the classical Brunn–Minkowski theorem [4].

THEOREM 1.10. *Let C_0, C_1 be non-empty convex sets in $\mathbf{R}^n$, and let*

$$C_\lambda = \lambda C_1 + (1-\lambda)C_0, \qquad 0 \leqslant \lambda \leqslant 1.$$

Denote by $|C|$ the n-dimensional Lebesgue measure of C. Then

$$|C_\lambda|^{1/n} \geqslant \lambda |C_1|^{1/n} + (1-\lambda)|C_0|^{1/n}.$$

REMARK. If $C_0 = \{0\}$, then $C_\lambda = \lambda C_1$.

In the case of Gaussian measures we have the following.

THEOREM 1.11. *Let C_0, C_1 and C_λ be as in Theorem 1.10, and let A be a real, positive-definite, n-square matrix. Let*

$$\mu_G(C) = \int_C \exp[-(x, Ax)/2]\,dx.$$

Then

$$\mu_G(C_\lambda) \geqslant \mu_G(C_1)^\lambda \mu_G(C_0)^{1-\lambda}.$$

Proof. Define the convex set

$$D = \{(\lambda, x) | 0 \leqslant \lambda \leqslant 1,\ x \in C_\lambda\} \subset \mathbf{R} \times \mathbf{R}^n.$$

Then

$$\mu_G(C_\lambda) = \int_{\mathbf{R}} \chi_D(\lambda, x) \exp[-(x, Ax)/2]\,dx.$$

Since the integrand is log concave in (λ, x), $\mu_G(C_\lambda)$ is log concave by Theorem 1.1.

Q.E.D.

As a corollary to Theorem 1.11 we quote a theorem of L. Gross [5]. The Gaussian measure μ_n on $\mathbf{R}^n$ defines the measure of a Borel set $B \subset \mathbf{R}^n$ to be

$$\mu_n(B) = (2\pi)^{-n/2} \int_B \exp[-(x, x)/2]\, dx.$$

THEOREM 1.12. (L. Gross) *Let C be a convex, balanced set in $\mathbf{R}^n \times \mathbf{R}^m$, let D be the intersection of C with $\mathbf{R}^m$, and let E be the projection of C on $\mathbf{R}^n$. Then*

$$\mu_{n+m}(C) \leq \mu_n(E)\mu_m(D).$$

Proof. Let C_x be the intersection of C with the plane parallel to $\mathbf{R}^m$ through $x \in \mathbf{R}^n$; in particular, $C_0 = D$. By the symmetry of C and Theorem 1.11, $x \to \mu_m(C_x)$ is log concave and even on $\mathbf{R}^n$, and hence $\mu_m(C_x)$ is maximal for $x = 0$. Thus

$$\mu_{n+m}(C) = \int_E \mu_m(C_x)\, d\mu_n(x) \leq \mu_m(D) \int_E d\mu_n(x) = \mu_n(E)\mu_m(D).$$

Q.E.D.

Let us return to the Brunn-Minkowski theorem (1.11) for Gaussians. By passing to the limit $n \to \infty$, the same theorem obviously remains true for infinite-dimensional Gaussian measures, for example, the Wiener measure. In that case we should deal with measurable, convex sets of Wiener paths. We shall consider here particular convex sets of paths, namely those passing through a convex set $C_\lambda \subset \mathbf{R}^n$ for all t.

With C_λ, $0 \leq \lambda \leq 1$, defined as in Theorem 1.10, consider the fundamental solution $G_\lambda(x, y; t)$, $t \geq 0$, of the diffusion equation with potential V, in $\mathbf{R}^n$, defined by

$$\left(\frac{\partial}{\partial t} - \frac{1}{2}\Delta_x + V(x)\right) G_\lambda(x, y; t) = 0;$$

$$G_\lambda(x, y; 0) = \delta(x - y), \qquad x, y \in C_\lambda;$$

$$G_\lambda(x, y; t) = 0, \qquad x \in \partial C_\lambda, \qquad \forall y, t;$$

$$\equiv 0, \qquad x \notin C_\lambda \quad \text{or} \quad y \notin C_\lambda.$$

THEOREM 1.13. *Let $V(x)$ be a convex function. Then $G_\lambda(x, y; t)$ is log concave in $(x, y, \lambda) \in \mathbf{R}^n \times \mathbf{R}^n \times [0, 1]$.*

Proof. Use the Trotter product formula with $x_0 = x$, $x_N = y$:

$$G_\lambda(x, y; t) = \lim_{N \to \infty} (2\pi t/N)^{-nN/2} \int_{\mathbf{R}^{n(N-1)}} dx_1 \ldots dx_{N-1} \times$$
$$\times \prod_{j=1}^{N} \left\{ \exp\left[-\frac{N}{2t}(x_j - x_{j-1})^2 - \frac{t}{N} V(x_j) \right] \chi_{C_\lambda}(x_j) \right\}.$$

The integrand is log concave in $(x, x_1, \ldots, x_{N-1}, y, \lambda)$. Finally the pointwise limit of a sequence of log concave functions is log concave.

Q.E.D.

COROLLARY 1.14. *In addition to the hypotheses of Theorem 1.13, either let* C_0 *and* C_1 *be compact or let* $\exp(-tV)$ *be in* $L^1(\mathbf{R}^n)$, $\forall t > 0$. *Define*

$$Z_\lambda(t) = \int_{C_\lambda} G_\lambda(x, x; t)\, dx = \operatorname{tr} e^{-tH},$$

with $H = -\frac{1}{2}\Delta + V$. *Then* $Z_\lambda(t) < \infty$ *and* $Z_\lambda(t)$ *is log concave in* λ.

Proof. That $Z_\lambda(t)$ is finite is a standard result and can be proved from the Trotter product formula above using Hoelder's inequality. The log concavity of $Z_\lambda(t)$ follows from Theorems 1.1 and 1.13.

Q.E.D.

COROLLARY 1.15. *Let* $V(x)$ *and* C_λ *be as in Corollary* 1.14. *Let* $\varepsilon_0(\lambda)$ *be the lowest eigenvalue of the equation*

$$[-\tfrac{1}{2}\Delta + V(x)]f(x) = \varepsilon_0(\lambda) f(x),$$

with $f(x) = 0$ *for* $x \in \partial C_\lambda$. *Then* $\varepsilon_0(\lambda)$ *is a convex function of* $\lambda \in [0, 1]$.

Proof. Since e^{-tH} is trace class, $Z_\lambda = \sum_{j=0}^{\infty} \exp[-t\varepsilon_j(\lambda)]$, $\varepsilon_{j+1}(\lambda) \geqslant \varepsilon_j(\lambda)$ and each $\varepsilon_j(\lambda)$ has finite multiplicity. Then

$$\varepsilon_0(\lambda) = -\lim_{t\to\infty} t^{-1} \ln Z_\lambda(t),$$

and, since the pointwise limit of a sequence of convex functions is convex, Corollary 1.15 is proved.

Q.E.D.

1.4. The one-dimensional plasma

In this section we apply the previous theorems to an old problem in physics, namely, to the one-dimensional, one-component plasma in a neutralizing background. We shall consider both the classical and quantum-mechanical cases. The latter requires the introduction of the Wiener integral, and thus provides another example of the application of our theorems to functional integrals. The object of our investigations is to show that long-range order exists for sufficiently large coupling constant, i.e. that the one-particle distribution function is a non-constant periodic function. The occurrence of this phenomenon was first predicted by Wigner [6].

Let $\boldsymbol{x} = (x_{-n}, \ldots, x_n)$ be the coordinates of $(2n+1)$ one-dimensional particles, each having a negative charge of one unit. The one-dimensional Coulomb potential between two unit charges separated by a distance $|x|$ is $-|x|$. Then the total potential energy of $(2n+1)$ particles in a 'box' $[-L, L]$ with a fixed uniform positive charge background of density ρ is

$$\phi(\boldsymbol{x}) = -\sum_{-n \leqslant i < j \leqslant n} |x_i - x_j| + \rho \sum_{i=-n}^{n} \int_{-L}^{L} |x_i - x| \, dx - \frac{\rho^2}{2} \int_{-L}^{L} \int_{-L}^{L} |x - y| \, dx \, dy. \tag{1.2}$$

We shall further assume that the total charge is zero, i.e.

$$2L\rho = 2n + 1.$$

Since ϕ is symmetric in the x_i, it is sufficient to consider the convex domain

$$C = \{\boldsymbol{x} | -L \leqslant x_{-n} \leqslant x_{-n+1} \ldots \leqslant x_n \leqslant L\}. \tag{1.3}$$

In C,

$$\phi(\boldsymbol{x}) = \rho \sum_{j=-n}^{n} \left(x_j - \frac{j}{\rho}\right)^2, \tag{1.4}$$

where a constant term in the potential has been neglected. Our methods are capable of handling the domain C as it stands, but then the function we wish to calculate, $\rho(x)$, will not be strictly periodic, except in the thermodynamic limit $n \to \infty$, $\rho =$ constant. To circumvent this difficulty we extend C to the larger convex domain

$$D = \{x | x_{-n} \leqslant x_{-n+1} \ldots \leqslant x_n \leqslant x_{-n} + 2L\}. \tag{1.5}$$

The domain D no longer confines the particles to the box, and we shall cheat a little by supposing that the expression (1.4) for ϕ is valid in all of D. The original walls of the box are still visible in ϕ.

Remark then that the domain D and the potential ϕ are invariant under the linear transformations R(reflection) and T(translation), defined by

$$(\mathrm{R}\mathbf{x})_j = -x_{-j} \tag{1.6}$$

$$(\mathrm{T}\mathbf{x})_j = x_{j+1} - \frac{1}{\rho}, \qquad -n \leqslant j \leqslant n-1;$$

$$(\mathrm{T}\mathbf{x})_n = x_{-n} + \frac{2n}{\rho}. \tag{1.7}$$

1.4.1. The classical case

The Gibbs distribution function of the jth particle is defined by (β is the reciprocal temperature)

$$\rho_j^{(n)}(x) = \frac{\int_D \delta(x - x_j) \exp[-\beta\phi(\mathbf{x})] \, d\mathbf{x}}{\int_D \exp[-\beta\phi(\mathbf{x})] \, d\mathbf{x}}.$$

The symmetry properties (1.6) and (1.7) imply that

$$\rho_0^{(n)}(x) = \rho_0^{(n)}(-x); \tag{1.8}$$

$$\rho_j^{(n)}(x) = \rho_0^{(n)}\left(x - \frac{j}{\rho}\right), \quad -n \leq j \leq n. \tag{1.9}$$

Since D is a convex domain in $\mathbf{R}^{2n+1}$, direct application of Theorem 1.9 gives that

$$\rho_0^{(n)}(x) = \exp(-\beta\rho x^2) F^{(n)}(x), \tag{1.10}$$

where $F^{(n)}$ is log concave; by eqn (1.8), $F^{(n)}(x)$ is also even.

We shall not go into the details of the existence of the limiting distribution functions

$$\rho_j(x) \equiv \lim_{n\to\infty} \rho_j^{(n)}(x)$$

in the thermodynamic limit ($n \to \infty$, $L \to \infty$, $2n+1 = 2L\rho$). Obviously, properties (1.8)–(1.10) remain true in the limit. It is also fairly clear, that the use of domain C instead of domain D would give the same distribution functions in the limit.

Thus far we have established part (*i*) of the following theorem:

THEOREM 1.16. (*i*) *The one-particle distribution functions of the classical one-dimensional plasma computed in D satisfy*

$$\rho_0(x) = \rho_0(-x),$$

$$\rho_j(x) = \rho_0\left(x - \frac{j}{\rho}\right),$$

$$\rho_0(x) = \exp[-\beta\rho x^2] F(x),$$

where $F(x)$ is a log concave, even function.

(*ii*) $$\int_{\mathbf{R}} |x|^\alpha \rho_0(x)\, dx \leq \left(\frac{\pi}{\beta\rho}\right)^{-\frac{1}{2}} \int_{\mathbf{R}} |x|^\alpha \exp[-\beta\rho x^2]\, dx$$

for $\alpha \geq 1$.

(*iii*) *For large values of β/ρ, the total density*

$$\rho(x) = \sum_{j=-\infty}^{\infty} \rho_j(x)$$

is non-trivially periodic.

Proof. (*ii*) Follows from Theorem 1.8, (*iii*) will be proved in § 1.4.3, Theorem 1.18.

REMARKS. (*i*) Theorem 1.16 (*ii*) states that the moments of the one-particle distribution functions are smaller than they would be without the restriction $x_j \leqslant x_{j+1}$.

(*ii*) The interpretation of Theorem 1.16 (*iii*) is that the plasma is in a crystalline state. The specific position of the crystal is a consequence of the hard walls that were imposed at $\pm L = \pm(n+\frac{1}{2})\rho$. This fact is reflected not only in the domain D (eqn (1.5)) but also in the expressions (1.4) for $\phi(\mathbf{x})$. Hard walls at positions $\pm L + \delta$ would translate the crystal through a distance δ.

(*iii*) The fact that $\rho(x)$ is not a constant has recently been proved by Kunz [7] who, by other methods, showed that to be true for all β,ρ except possibly for a countable number of values of β/ρ.

1.4.2. The quantum-mechanical case

The quantum-mechanical Hamiltonian of the system defined by equation (1.2), with $\hbar^2/m = 1$, is

$$H = -\tfrac{1}{2}\Delta + \phi(\mathbf{x}),$$

where

$$\Delta = \sum_{j=-n}^{n} \frac{\partial^2}{\partial x_j^2}.$$

We consider the case that the particles are spinless fermions. This means that H acts on square integrable, antisymmetric functions. As is well known, an equivalent statement in one dimension is that H acts on square integrable functions defined on $E = \{\mathbf{x} | x_{-n} \leqslant x_{-n+1}, \ldots, \leqslant x_n\}$ which vanish on the boundary of E. The 'box' condition requires that the functions vanish on the boundary of $C \subset E$. As in the classical case, we shall use the larger domain D instead of C.

The distribution function of the jth particle is then

$$\rho_j^{(n)}(x) = \frac{\mathrm{tr}_D\, \mathrm{e}^{-\beta H}\delta_j(x)}{\mathrm{tr}_D\, \mathrm{e}^{-\beta H}},$$

where $\delta_j(x)$ is the operator of multiplication by $\delta(x - x_j)$. Since H and D are invariant under the transformations R and T (eqns (1.6), (1.7)), the distribution functions again have the symmetry properties ((1.8), (1.9)).

To find the analogue of eqn (1.10), use the Trotter product formula for $\exp(-\beta H)$, which gives, with $\mathbf{x}^0 = \mathbf{x}^N$,

$$\mathrm{Tr}_D\, \mathrm{e}^{-\beta H}\delta_0(x) = \lim_{N\to\infty}\left(\frac{2\pi\beta}{N}\right)^{-(2n+1)N/2} \int_{D^N} \mathrm{d}\mathbf{x}^1 \ldots \mathrm{d}\mathbf{x}^N \times$$

$$\times \prod_{k=1}^{N} \exp\left[-\frac{N}{2\beta}\sum_{j=-n}^{n}(x_j^k - x_j^{k-1})^2 - \frac{\beta\rho}{N}\sum_{j=-n}^{n}\left(x_j^k - \frac{j}{\rho}\right)^2\right]\delta(x_0^1 - x). \qquad (1.11)$$

Since D^N is convex in $\mathbf{R}^{(2n+1)N}$, we can apply Theorem 1.9 to conclude that

$$\rho_0^{(n)}(x) = \exp(-\gamma x^2) H^{(n)}(x), \tag{1.12}$$

where $H^{(n)}$ is log concave, and where $\exp(-\gamma x^2)$ is, up to multiplication by an x-independent constant, what eqn (1.11) would give if D were replaced by $\mathbf{R}^{2n+1}$. But in that case the integrations separate into $(2n+1)$ independent integrations over $\mathbf{R}^N$. Therefore, $\exp(-\gamma x^2)$ is proportional to $G(x, x; \beta)$, where $G(x, y; t)$ is the fundamental solution (Green's function) of the differential equation, for $t > 0$,

$$\left(\frac{\partial}{\partial t} - \frac{1}{2}\frac{\partial^2}{\partial x^2} + \rho x^2\right) G(x, y; t) = 0;$$

$$G(x, y; 0) = \delta(x - y).$$

Using the well-known expression [8] for G, we obtain

$$\gamma = (2\rho)^{\frac{1}{2}} \tanh \beta(\rho/2)^{\frac{1}{2}}. \tag{1.13}$$

The analogue of Theorem 1.16 is now immediate.

THEOREM 1.17: *Theorem 1.16 is correct for the quantum-mechanical plasma of spinless fermions except that in part (ii) $\beta\rho$ is replaced by γ (eqn (1.13)) and in part (iii) β/ρ is replaced by γ/ρ^2.*

Remarks (*i*) and (*ii*) after Theorem 1.16 also apply here.

We turn next to the demonstration that parts (*iii*) of Theorems 1.16 and 1.17 follow from parts (*i*) of those theorems.

1.4.3. Can modified theta functions be constant?

Let $f(x) = \exp(-\lambda x^2) F(x)$ with $F(x)$ even and log concave and $\lambda > 0$. Consider

$$\rho(x) = \sum_{j=-\infty}^{\infty} f(x - j). \tag{1.14}$$

The question to which we address ourselves here is whether or not F can be chosen so that $\rho(x)$ is constant. The answer, surprisingly, depends on λ. As Theorem 1.18 shows, $\rho(x)$ cannot be constant when λ is large, and thus parts (*iii*) of Theorems 1.16 and 1.17 are proved.

Define the Fourier transform by

$$\hat{f}(k) = \int_{\mathbf{R}} e^{2\pi i k x} f(x)\, dx. \tag{1.15}$$

Then, by the Poisson summation formula,

$$\rho(x) = \sum_{j=-\infty}^{\infty} f(j)\, e^{-2\pi i j x}. \tag{1.16}$$

Therefore $\rho(x)$ is constant iff $\hat{f}(j) = 0$ for $j = \pm 1, \pm 2, \ldots$.

THEOREM 1.18. *Let $\rho(x)$ be defined as in eqn* (1.14). *Then there exists a λ_0, $0<\lambda_0<\infty$,* such that

(a) *For all $\lambda>\lambda_0$ and for all F even and log concave, $\rho(x)$ is not constant*;

(b) *For all $\lambda<\lambda_0$, $\lambda>0$ there exists an even, log concave F such that $\rho(x)=$ constant.*

Proof. (*i*) Existence of λ_0: If, for some λ, there is an $F(x)$ that leads to a constant $\rho(x)$, then, for $\mu<\lambda$, the log concave function $F(x)\exp[(\mu-\lambda)x^2]$ gives the same constant $\rho(x)$.

(*ii*) $\lambda_0<\infty$: Normalize to $F(0)=1$. Then $\rho(0)\geqslant 1$, and

$$\rho(\tfrac{1}{2})\leqslant \sum_{j=-\infty}^{\infty} \exp[-\lambda(j-\tfrac{1}{2})^2]\leqslant 2\,\mathrm{e}^{-\lambda/4}\sum_{j=0}^{\infty}\mathrm{e}^{-2\lambda j}=2\,\mathrm{e}^{-\lambda/4}(1-\mathrm{e}^{-2\lambda})^{-1}.$$

This gives the simple estimation $\lambda_0<3$.

(*iii*) $\lambda_0>0$: We indicate how to construct an example of constant ρ for λ sufficiently small. Choose a non-constant, even, log concave function G, and normalize it so that $g(x)\equiv\exp(-\lambda x^2)G(x)$ satisfies

$$\textstyle\int_{\mathbf{R}} g(x)\,dx=\hat{g}(0)=1.$$

Define

$$\hat{f}(k)=\prod_{j=1}^{\infty}\hat{g}(k/j), \tag{1.17}$$

which is the Fourier transform of the convolution

$$f(x)=\prod_{j=1}^{\infty}{}^{*}j\exp(-\lambda j^2x^2)G(jx). \tag{1.18}$$

The infinite product (1.17) is defined and $\hat{g}(k)>0$ in a neighbourhood of $k=0$, since

$$1>\hat{g}(k/j)\approx 1+\tfrac{1}{2}(k/j)^2\hat{g}''(0)>0 \quad \text{for} \quad |k/j|\ll 1.$$

Equation (1.18) then follows from the Lebesgue dominated convergence theorem, and $f\not\equiv 0$. Now Theorem 1.9 applied to eqn (1.18) gives

$$f(x)=\exp(-\alpha\lambda x^2)F(x),$$

where $F(x)$ is log concave and even and $\alpha=(\sum_{j=1}^{\infty}j^{-2})^{-1}=6\pi^{-2}$.

It is now sufficient to determine λ and G such that $\hat{g}(\pm 1)=0$; then, by eqn (1.17),

$$\hat{f}(j)=0 \quad \text{for all integers} \quad j\neq 0,$$

and we are done. Take $G(x) = [1 + \sqrt{(\lambda/\pi)}]\chi(x)$, where χ is the characteristic function of $[-\frac{3}{4}, \frac{3}{4}]$. Then

$$\lim_{\lambda \to 0} \hat{g}(k) = (\pi k)^{-1} \sin \tfrac{3}{2}\pi k; \lim_{\lambda \to \infty} \hat{g}(k) = 1.$$

Therefore λ can be chosen such that $\hat{g}(\pm 1) = 0$.

Q.E.D.

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Note

Since this work was completed we have found that Theorem 1.1 has been proved independently by A. Prekopa, *Acta Math. Szeged* **32**, 301–15 (1971), and Y. Rinott, Thesis, Weizmann Institute, Rehovoth, Israel (Nov. 1973).